



Duration: 3hrs

Max Marks:80

- N.B. : (1) Question No 1 is Compulsory.
 (2) Attempt any **three** questions out of the remaining **five**.
 (3) All questions carry equal marks.
 (4) Assume suitable data, if required and state it clearly.

Q1. Solve any **four** from following. [20]

- What are the issues in Machine learning?
- Explain Regression line, Scatter plot, Error in prediction and Best fitting line.
- Explain the concept of margin and support vector.
- Explain following performance metrics along with an example Accuracy, Precision, Recall and F1 score.
- Explain Logistic Regression

Q2. a. Explain the steps of developing Machine Learning applications. [10]

b. Write short note on Linear Discriminant projection along with an example. [10]

Q3. a. Demonstrate CART method along with an example. [10]

b. Following table shows the midterm and final exam grades obtained for students in a database course. Using linear regression to predict the final exam grade of a student who received 86 in the midterm exam. [10]

Midterm exam (X)	72	50	81	74	94	86	59	83	86	33	88	51
Final exam (Y)	84	53	77	78	90	75	49	79	77	52	74	90

Q4. a. Explain the Random Forest algorithm in detail. [10]

b. Explain the different ways to combine the classifiers. [10]

Q5. a. Describe Multiclass classification. [10]

b. Demonstrate MST algorithm along with example. [10]

Q6. Write detailed note on following. (Any two) [20]

- Performance Metrics for Classification
- Principal Component Analysis for Dimension Reduction
- DBSCAN algorithm.

Q p code
85954

Prog. code
1T00737

Duration: 3 Hours

[Max. Marks: 80]

- N.B.: (1) Question No 1 is Compulsory.
 (2) Attempt any **THREE** questions out of the remaining **FIVE**.
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- Q1** Attempt any **FOUR** [20]
 A Explain Training error and Generalization error.
 B Differentiate between Supervised and unsupervised Learning
 C Differentiate between Linear regression and Logistic regression.
 D Explain issues in Machine learning.
 E Explain performance evaluation metrics for the classification.
- Q2** **A** Demonstrate MST algorithm along with example. [10]
 B Explain Logistics regression and performance evaluation metrics. [10]
- Q3** A Demonstrate steps to design a Machine Learning application. [10]
 B What is over fitting, under fitting and Bias variance trade-off with reference to Machine learning? [10]
- Q4** **A** Demonstrate Ensemble learning based Random Forest algorithm in detail. [10]
B Suppose we want Gini index to decide whether the car will be stolen or not. The target classification is "car is stolen?" which can be Yes or No, create a decision tree for the given data below. [10]

Car no	Colour	Type	Origin	Stolen ?
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Domestic	No
5	Yellow	Sports	Imported	Yes

Car no	Colour	Type	Origin	Stolen ?
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Domestic	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes

- Q5** **A** Give steps to design PCA dimensional reduction technique along with an example. [10]
 B Demonstrate DBSCAN algorithm along with example. [10]
- Q6** Write detailed note on following. (Any TWO) [20]
A Write a short note on XGBoost ensemble method.
 B Explain support vector machine as constraint optimization problem.
 C SVM Kernel trick

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- N.B. :** (1) Question No 1 is Compulsory.
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- Q1 Attempt any **FOUR** from the following [20]
 A Explain any five business applications of Machine learning.
 B What is dimensionality reduction? Explain how it can be utilized for classification and clustering task in Machine learning.
 C Explain performance evaluation metrics for binary classification with suitable example.
 D Explain Gini index along with an example.
 E Explain the concept of k fold cross validation.

- Q2 A Write a short note on issues in Machine Learning. [10]
 B Compare Bagging and Boosting with reference to ensemble learning. Explain how these methods help to improve the performance of the machine learning model. [10]

- Q3 A Consider the example below where the mass, y (grams), of a chemical is related to the time, x (seconds), for which the chemical reaction has been taking place according to the table. Find the equation of the regression line. Also explain performance evaluation measures for regression. [10]

Time, x (seconds)	5	7	12	16	20
Mass, y (grams)	40	120	180	210	240

- B What is Density based clustering? Explain the steps used for clustering task using Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm. [10]
 Q4 A Explain Clustering with minimal spanning tree along with example. [10]
 B Consider the dataset given below with 3 features Color, Wig, Num. Ears and one output variable Emotion [10]

Color	G	G	G	B	B	R	R	R	R
Wig	Y	N	N	N	N	N	N	N	Y
Num. Ears	2	2	2	2	2	2	2	2	3
Emotion	S	S	S	S	H	H	H	H	H

- i) Find root node of decision tree using GINI index
 ii) Explain techniques can be used to handle over fitting in decision trees?
 Q5 A Consider the use case of Email spam detection. Identify and explain the suitable machine learning technique for this task. [10]
 B Explain the Dimensionality reduction technique Linear Discriminant Analysis and its real-world applications. [10]
 Q6 A Define following terminologies with reference to Support vector machine: Hyper plane, Support Vectors, Hard Margin, Soft Margin, Kernel [10]
 B Explain Ensemble learning algorithm Random Forest and its use cases in real world applications. [10]

Time: 3hrs

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[Total Marks:80]

Q1 Attempt any **FOUR** from the following [20]

- A Explain how to choose the right algorithm for machine learning application.
- B Explain Linear Discriminant Analysis.
- C Explain any five performance measures along with example.
- D Differentiate between Logistic regression and Support vector machine.
- E Explain the following Receiver operating characteristics curve and Area under curve.

Q2 A Explain clustering with minimal spanning tree with reference to Graph based clustering. [10]

B Explain the terms overfitting, underfitting, bias & variance tradeoff w.r.t. Machine Learning. [10]

Q3 A Explain the concept of regression and enlist its types. A clinical trial gave the data for BMI and Cholesterol level for 10 Patients as shown in table below. Identify the machine learning method used to solve the above problem and predict the likely value of Cholesterol level for someone who has BMI of 27. [10]

BMI	17	21	24	28	14	16	19	22	15	18
Cholesterol	140	189	210	240	130	100	135	166	130	170

B Explain the necessity of cross validation in Machine learning applications and K-fold cross validation in detail. [10]

Q4 A Explain **support vector machine as a constrained optimization** problem. [10]

B Explain the concept of decision tree. Consider the dataset given in a table below. The dataset has 3 features as Past Trend, Open interest, Trading volume and one class label as Return. Compute the Gini Index for all features and specify which node will be chosen as a root node in decision tree. [10]

Past Trend	Open Interest	Trading Volume	Return
Positive	Low	High	Up
Negative	High	Low	Down
Positive	Low	High	Up
Positive	High	High	Up
Negative	Low	High	Down
Positive	Low	Low	Down
Negative	High	High	Down
Negative	Low	High	Down
Positive	Low	Low	Down
Positive	High	High	Up

Q5 A Explain kernel Trick in support vector machine. [10]

B Explain different ways to combine classifiers. [10]

Q6 Write any **TWO** from the following [20]

- A Explain multiclass classification techniques.
- B Explain in detail Principal Component Analysis for Dimensionality reduction
- C Explain DBSCAN algorithm along with example

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Q1 Attempt any **four** from following. [20]

- A** How to choose the right ML algorithm?
B Explain Regression line, Scatter plot, Error in prediction and Best fitting line.
C Explain the concept of feature selection and extraction.
D Explain K-means algorithm.
E Explain the concept of Logistic Regression

Q2 **A** Explain any five applications of Machine Learning. [10]

B Explain Multivariate Linear regression method. [10]

Q3 **A** Create a decision tree using Gini Index to classify following dataset for profit. [10]

Age	Competition	Type	Profit
old	Yes	software	down
old	No	software	Down
old	No	hardware	Down
mid	Yes	software	Down
mid	Yes	hardware	Down
mid	No	hardware	Up
mid	No	software	Up
new	Yes	software	Up
new	No	hardware	Up
new	no	software	Up

B Find SVD for $A = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}$ [10]

Q4 **A** Explain the Random Forest algorithm in detail. [10]

B Explain the concept of bagging and boosting. [10]

Q5 **A** Describe Multiclass classification. [10]

B Explain the concept of Expectation Maximization Algorithm. [10]

Q6 Write detailed note on following. (**Any two**) [20]

- A** Linear Regression
B Linear Discriminant Analysis for Dimension Reduction
C DBSCAN

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Q1. Solve any **four** from following. [20]

- What are the issues in Machine learning?
- Explain Regression line, Scatter plot, Error in prediction and Best fitting line.
- Explain the concept of margin and support vector.
- Explain the distance metrics used in clustering.
- Explain Logistic Regression

Q2. a. Explain the steps of developing Machine Learning applications. [10]

b. Explain Linear regression along with an example. [10]

Q3. a. Create a decision tree using Gini Index to classify following dataset. [10]

Sr. No.	Income	Age	Own Car
1	Very High	Young	Yes
2	High	Medium	Yes
3	Low	Young	No
4	High	Medium	Yes
5	Very High	Medium	Yes
6	Medium	Young	Yes
7	High	Old	Yes
8	Medium	Medium	No
9	Low	Medium	No
10	Low	Old	No
11	High	Young	Yes
12	Medium	Old	No

b. Describe Multiclass classification. [10]

Q4. a. Explain the Random Forest algorithm in detail. [10]

b. Explain the different ways to combine the classifiers. [10]

Q5. a. Compute the Linear Discriminant projection for the following two-dimensional dataset. $X_1 = (x_1, x_2) = \{(4,1), (2,4), (2,3), (3,6), (4,4)\}$ and $X_2 = (x_1, x_2) = \{(9,10), (6,8), (9,5), (8,7), (10,8)\}$ [10]

b. Explain EM algorithm. [10]

Q6. Write detailed note on following. (Any two) [20]

- Performance Metrics for Classification
- Principal Component Analysis for Dimension Reduction
- DBSCAN
